

Enhancing Object Detection Robustness in Low-Light Conditions via Self-Calibrated Illumination

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Abstract—Object detectors achieve high accuracy on clear images but experience severe performance drops in low-light environments. This project investigates state-of-the-art methods for improving detection under degraded illumination. Utilizing publicly available official code, we will reproduce the CVPR 2022 paper “Toward Fast, Flexible, and Robust Low-Light Image Enhancement” (SCI). We adapt this pre-trained enhancement model to a new application domain by integrating it as a preprocessing step for the state-of-the-art YOLOv8 architecture, evaluating its performance on the Exclusively Dark (ExDark) object detection dataset. Our report details data ingestion, augmentation strategies (applying synthetic noise to test robustness), and a comparative evaluation using mean Average Precision (mAP). By leveraging this available code, we aim to demonstrate how preprocessing dark images can recover object detection accuracy in modern, real-time pipelines.

Index Terms—Computer Vision, Object Detection, Low-Light Enhancement, YOLO, Deep Learning

I. INTRODUCTION

Object detection is crucial in many real-world applications such as autonomous driving and surveillance. However, degraded image quality, specifically low illumination at night, significantly hampers detector performance. Standard detectors often see dramatic drops in mean Average Precision (mAP) under such conditions. A detector that scores high on clean, well-lit images might drop by tens of points when images are dark.

To address this, our project follows the “Reproduce and Adapt” methodology. We will reproduce the state-of-the-art low-light enhancement method “SCI” (CVPR 2022) using the authors’ publicly available codebase. We adapt it to a new application domain by deploying it on the ExDark dataset, which contains exclusively low-light images across multiple object classes. Rather than training a new object detector from scratch, we will treat the SCI model as an advanced preprocessing step. We will evaluate how much this preprocessing improves the mAP of a modern, pre-trained real-time detector (YOLOv8) compared to running YOLOv8 on the raw, unenhanced dark images.

II. LITERATURE REVIEW

Recent research addresses object detection under image degradations through three primary strategies: (1) Image enhancement to reconstruct inputs prior to detection, (2) Joint adaptation and unified frameworks that bridge the gap between enhancement and detection, and (3) Data augmentation and specialized detection models designed for inherent robustness.

A. Enhancement via Neural Networks

A prominent direction in computer vision is to process degraded images to make them more detector-friendly before they enter the downstream detection pipeline. Early methods like Zero-DCE [2] revolutionized this space by formulating low-light enhancement as a task of image-specific curve estimation. Instead of relying on paired datasets (dark images matched perfectly with bright images), Zero-DCE uses a deep neural network to estimate pixel-wise high-order curves, allowing for zero-reference training guided by spatial consistency and exposure control losses.

Subsequent works have advanced this foundational concept by tackling the complex physics of lighting. For instance, URetinex-Net [3] integrates a deep unfolding network based on classical Retinex theory. By explicitly decomposing an image into its fundamental illumination and reflectance maps, the model can systematically handle severe low-light degradation while preserving color fidelity. Similarly, Xu et al. [4] addressed the critical issue of signal-to-noise ratio (SNR) in dark environments. Because low light artificially amplifies camera sensor noise, their SNR-aware enhancement method dynamically assesses the spatial noise distribution, selectively boosting illumination in dark areas without amplifying the underlying visual grain. Recently, complex transformer architectures have been adapted for this task. Retinexformer [5] introduces a one-stage Retinex-based transformer that utilizes self-attention mechanisms to capture long-range dependencies in the image, managing both illumination correction and noise suppression jointly.

While theoretically powerful, a major limitation of these complex generative models is that they introduce significant

computational latency, rendering them impractical for real-time video feeds. To address this critical bottleneck, Ma et al. introduced the Self-Calibrated Illumination (SCI) framework [1]. SCI employs a cascaded illumination learning process with a unique weight-sharing mechanism. During inference, this complex cascade reduces to a single, highly efficient operation. Unlike heavy generative models, SCI can be seamlessly integrated into downstream vision tasks like object detection, enhancing images in a fraction of a millisecond without introducing noticeable latency. This balance of speed and accuracy makes SCI the ideal preprocessing candidate for our methodology. Beyond traditional RGB processing, cutting-edge research is also exploring alternative hardware; for example, the RETINEV framework [6] utilizes event-based cameras, which possess higher dynamic ranges and bypass traditional motion blur, to estimate illumination in extreme darkness.

B. Joint Adaptation and Unified Frameworks

Another critical research track focuses on bridging the semantic gap between image processing and object detection. Standard object detectors are trained on well-lit datasets and can suffer severe performance drops if independent enhancement algorithms distort the underlying image features or introduce artifacts. To solve this, Wang et al. proposed HLA-Face [7], a joint high-low adaptation framework. HLA-Face ensures that both low-level features (such as color and texture) and high-level features (semantic representations) are aligned across normal and low-light domains, ensuring the detector remains highly accurate regardless of the input's lighting domain.

In the context of generic object detection, Liu et al. introduced Image-Adaptive YOLO (IA-YOLO) [8]. Instead of using a “black-box” neural network to generate a new image, IA-YOLO utilizes a small Convolutional Neural Network (CNN) to predict parameters for classical differentiable image processing (DIP) filters (e.g., gamma, contrast, and color balance). Because these filters are differentiable, the detection loss from the YOLO model can backpropagate through the entire network, allowing the system to dynamically adjust the image filters to optimize explicitly for object detection in adverse weather and lighting. Building on this streamlined concept, Ogino et al. [9] recently proposed ERUP-YOLO, which unifies these classical image filters into even simpler, differentiable operations, proving that task-driven adaptation can significantly enhance detection robustness without the computational overhead of heavy image reconstruction.

C. Data Augmentation, Datasets, and Real-Time Detectors

Beyond algorithmic preprocessing, data-centric approaches are vital for training inherently robust detectors. Sayed et al. [10] focused on the degradation of motion blur, demonstrating that a detector's Average Precision (AP) drops significantly when exposed to blurred objects. Rather than attempting to deblur the image, they augmented the training dataset with synthetically generated motion blur and dynamically

expanded the ground-truth bounding boxes to encompass the blur trajectories. By conditioning the models on blur-type, they effectively recovered performance, proving that smart data augmentation is a highly viable alternative to complex architectural changes.

Properly evaluating these diverse enhancement and augmentation methods requires dedicated, highly realistic datasets. While many researchers rely on synthetically darkened images, Loh et al. [11] introduced the Exclusively Dark (ExDark) dataset. ExDark provides a critical, real-world benchmark containing thousands of natural low-light images, ranging from twilight to extreme indoor darkness, complete with detailed object bounding boxes across multiple classes.

Historically, many of the aforementioned enhancement papers evaluated their downstream detection gains using older, two-stage architectures like Faster R-CNN, which are computationally heavy. There remains a notable gap in the literature regarding how modern, highly optimized, single-stage real-time detectors, such as YOLOv8 [12], respond to these specialized low-light enhancement models. YOLOv8 utilizes an advanced CSPNet backbone and an anchor-free head, achieving state-of-the-art speeds. By coupling the ultra-fast preprocessing capabilities of SCI [1] with the optimized YOLOv8 architecture on the challenging ExDark benchmark [11], our project directly addresses this gap. We aim to provide a modern, comprehensive comparative evaluation of these established techniques, demonstrating the viability of real-time low-light object detection for practical, edge-computing applications.

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